SANTANDER X · QUANTUM AI LEAP · 2026

EmpowerTours
Quantum Portfolio

Q-Day-resistant DeFi, shipped today.

QAOA on IBM Heron · Hedged post-quantum signatures · On-chain provenance on Monad

THE THREAT

Every ECDSA signature you publish today is a hostage to tomorrow's QPU.

This is not *harvest-now-decrypt-later — signatures aren't secret, and your orders are already public. It is worse in one specific way: the day the curve falls, every signature ever made with it becomes* ***forgeable****, so a genuine 2026 authorisation and one fabricated in 2035 are no longer distinguishable. Your audit trail stops proving anything. And on an EVM chain every transaction you have ever sent already exposes the public key that a QPU turns into your private key.*

WHY DEFI GETS HIT FIRST

Public chains are an open archive of signatures.

- Every transaction broadcasts an **ECDSA signature** in the clear.
- Indexers store them forever.
- A cryptographically-relevant QPU forges any one of them.
- The chain doesn't know which.

TODAY'S DEFENSE

None. Most wallets still sign with secp256k1.
Most "post-quantum DeFi" decks are roadmaps.

THE STACK

Three layers. Each one survives the other two.

1. ALLOCATE

QAOA on IBM Heron QPU picks Markowitz-optimal weights under risk constraints.

2. SIGN

Hedged PQ: every order carries ML-DSA-65 + SLH-DSA-SHAKE-256s + Ed25519. Any one survives.

3. ANCHOR

SHA-256(order) committed to Monad's `AuditAnchorV2` before the vault will execute it.

> *One layer broken ≠ system broken. The threat model is* every algorithm we trust today is provisionally trusted.

THE QUANTUM LAYER

QAOA on real hardware, reported with statistical honesty.

- **IBM Heron**, XY-ring-mixer QAOA, reps=3. Raw counts shipped — recompute it yourself.
- **Stocks** (`ibm_fez`, n=1): **13 → 39**, ×2.19 vs random, Fisher p = 0.00039 — the number we used to lead with.
- **Then we built the right null.** Those qubits are biased (P(1) = 0.41–0.59, not 0.375). Against a null from the run's own marginals the lift is **×1.26, p = 0.12**.
- **And replicated it.** 20 paired runs: no mitigation effect on P(opt|feasible), **p = 0.177**.
- **The simulator still shows real structure (×1.67).** The hardware never did.

WHAT THIS IS, HONESTLY

QAOA is **not** faster than the classical baseline at this size — that baseline is `NumPyMinimumEigensolver` (brute-force exact, `src/solvers.py:33`), which is optimal and instant on 8 assets.

We dismantled our own headline and published that. The lift was single-qubit readout bias measured against a null that assumed unbiased qubits. Finding it required attacking our own result — which is the contribution.

THE SIGNATURE LAYER

Don't bet on one algorithm. Hedge the bet.

ALGORITHM	FAMILY	WHY INCLUDE
`ML-DSA-65`	Lattice (FIPS 204)	NIST PQ standard. Fast.
`SLH-DSA-SHAKE-256s`	Hash (FIPS 205)	Different math. Slow but conservative.
`Ed25519`	Classical EC	Battle-tested. Hedge for "did we mis-port a new standard?"

> *If ML-DSA falls to a 2028 cryptanalysis paper, SLH-DSA still authenticates the order. If both lattice and hash fall, the classical signature carries it pre-Q-Day.*

THE PROVENANCE LAYER

A hash chain the vault refuses to disobey.

1. Agent builds order. PQ-signs it.
2. `AuditAnchorV2.anchor(orderHash, execCommitment, seq)` on Monad mainnet. Commits the order and the one execution it authorises.
3. `UniswapRoutingVault.executeAndRoute(orderHash, …)` recomputes the commitment from its own calldata and reverts unless it equals `ANCHOR.execCommitmentOf[msg.sender][orderHash]`. Anchoring an order is not enough — the trade must be the trade that was signed for.

WHAT THIS KILLS

- Replay (sequence enforced per wallet)
- Off-chain order tampering (hash doesn't match)
- Vault impersonation (pair allowlist immutable per deploy)
- Sandwich-DoS (`amountOutMin` from caller, not on-chain quote)

THE OBVIOUS OBJECTION

Yes — the chain transaction is still signed with ECDSA.

What we do NOT protect

A quantum computer that breaks ECDSA does not need to forge our order. It takes the wallet key and moves the funds directly.

We do not claim otherwise. `SECURITY.md` states this before anyone asks — it is the first item under What the code does not protect.

What we DO protect — today, no quantum computer required

- **The instruction.** The vault recomputes the commitment from its own calldata and reverts unless the trade is the trade that was signed for. Anchoring is not enough.
- **The audit trail past Q-Day.** Signatures over the order stay unforgeable even after ECDSA falls.
- **The upgrade path.** When chains add post-quantum transaction signatures, the layer above them already exists.

> *We protect the instruction, not the key. Key custody is the custodian's job — which is exactly why HSM custody gates our own managed-signing revenue rather than launching with it.*

PROOF, NOT PROMISES

Live on Monad MAINNET.
448 tests. ZK-verified.

4

CONTRACTS LIVE
MONADSCAN-VERIFIED

448

TESTS PASSING
267 PYTHON + 181 FOUNDRY

3

SIGNATURE ALGORITHMS
PER ORDER, HEDGED

AuditAnchorV2	0x8422b555dce11913a4657c2f47c839637fc71ffd
UniswapRoutingVault	0xcc60db5e123cb3150d5f11ca5526a79b4f31113f
MorphoSupplyAdapter	0x6d42fa32880add1d794abbf98c5cd104fe332d89
MLDSAAttestationV2	0xfeef24a5dbf43e9de8ac0d0eab0f0141e980a52c

Current set, deployed 2026-08-25, all Monadscan-verified. The demo on the next slide ran on the **superseded** pair and the v1 attestation, which are named there.

THE END-TO-END DEMO — REAL VALUE, ONE HASH

Executed on mainnet against the contracts live at the time. Three of the four were **superseded** on 2026-08-25 by the key-rotation migration and are marked below; the transactions remain on chain and remain valid evidence that the loop works.

QPU decision → PQ signature ZK-verified on-chain → yield.

STEP	CONTRACT	WHAT HAPPENED (MONAD MAINNET)
1. Attest	`MLDSAAttestationV2`	Order's **ML-DSA-65 signature verified on-chain via ZK proof** (1.19M gas measured, vs 8.1M measured for native EVM ML-DSA); `pqAttested = true`
2. Anchor	`AuditAnchorV2`	Hash `0xb4eceb58…3325` committed, immutable
3. Swap	`UniswapRoutingVault`	0.1 MON → **2,755 micro-USDC** (0.002685 USDC ≈ $0.002 — a deliberately tiny live-value demo) via live Uniswap v3, anchor-gated
4. Yield	`MorphoSupplyAdapter`	USDC supplied into a live **Morpho** market (~4.75% APY), non-custodial

> *The routing leg and the ZK attestation share one 32-byte `orderHash`; the yield leg carries its own signed order, because AuditAnchorV2 binds one commitment per order and each executor demands its own. A reviewer replays the events and verifies every step on-chain — without trusting us.*

WHY US

We ship the part everyone else handwaves.

- **Most PQ-DeFi pitches:** slides about a future migration.
- **Most QAOA-finance pitches:** one hand-picked seed on a simulator.
- **Most provenance pitches:** off-chain Merkle trees nobody verifies.
- **Ours:** verified contracts, real QPU runs with statistical rigor, on-chain hash chain enforced by the vault itself.
- One repo. One CI. Reproducible from a cold clone.

WHY NOW

The deadline for signatures is 31 December 2031.

EO 14412 — "Securing the Nation Against Advanced Cryptographic Attacks", 22 June 2026

OMB M-26-15 — five-phase migration schedule, 24 June 2026

DATE	REQUIREMENT
31 Dec 2030	post-quantum **encryption**
31 Dec 2031	post-quantum **authentication**

Authentication is signatures.

Everything we built sits on the **2031** line.

> *NIST IR 8547 independently deprecates RSA and ECC by 2030, disallows them by 2035.*

THE GAP

Everyone is protecting the key. Nobody proves the settlement.

WHO	BUILDING	LEAVES
Coinbase (23 Jul 2026)	quantum-resistant custody	the key at rest
BTQ + QBits	quantum-secure treasury, new chain	requires chain migration
PQShield · KETS · InfiniQuant	PQC primitives + hardware	components

Native EVM ML-DSA verification costs a measured **8.1M gas** (NIST-compliant) or **4.9M** (ZKNoxHQ's optimised ETHDilithium), both KAT-passing and deployed on Sepolia under the Ethereum Foundation's Kohaku project. Against Monad's 150M block that is includable — just expensive.

We measured 1.20M — 6.8x cheaper than the native NIST verifier, and 0.8% of one block. It ran on mainnet, and it needs no chain migration and no change to key storage.

> *Which makes them channel partners, not incumbents.*

REVENUE

The attestation is the billable unit.

LINE	PRICE
SDK licence per institution / yr	**$60–150k**
Per attestation settled	**$0.02–0.10**
Managed cosigner hosted, HSM	monthly

Atomic. Countable on-chain by both parties. Scales with the customer's volume, not our headcount.

> *Hosted signing ships* ***after*** *the audit and HSM custody — not before. Selling it off chmod-600 key files would be malpractice.*

These prices are proposals, not observed. We have sold nothing.

MARKET

We size bottom-up, because top-down is noise.

2026 digital-asset custody "market size", by vendor report:

`$0.7T · $793B · $834B · $954B · $1.05T`

A 50% spread means they are measuring different things. We will not quote the largest one at you.

```
N institutions × $100k licence
+ attestations × $0.05

N=250, full          → $25M ARR
N=250, 4% (10)       → $1.0M ARR
```

We do not have a defensible N.

Establishing it is the first thing funding buys — and it is a question with a knowable answer.

TRACTION

Zero customers.
Zero revenue. No LOIs.

No conversation with any institution has happened. We would rather be marked down for an empty pipeline than imply one we do not have.

What does exist, and is checkable:

- 4 contracts live on Monad **mainnet**, Monadscan-verified
- one end-to-end run with **real value**
- **448 tests**, 0 skipped with the fork env set
- 2 IBM Heron runs, job IDs + raw counts published
- every documented reviewer command **run and reproducible**

GO-TO-MARKET

One design partner, then two channels.

0–3 months — one custodian or tokenisation platform. Unpaid, for a public case study. Establishes N and finds whether compliance or engineering holds the budget.

3–9 months — channel through the PQC vendors. They sell into our exact buyer and have no settlement story.

6–12 months — chain-level distribution. Monad and peers have a direct interest in being where PQ settlement is cheapest.

> *Gate on the design partner before spending on either channel. If nobody will sign, the hypothesis is wrong — and we learn that in twelve weeks, not twelve months.*

TEAM

Two people.
One ships it, one runs it.

Earvin Gallardo Bravo — Founder, engineering

Wrote the full stack: the Solidity contracts, the SP1 ZK guest program, the hedged PQ signing layer, the QAOA hardware path and the app. Every commit in this repository.

Brisa Mar Hernández Hernández — Operations

Runs everything outside the codebase — company, operations and delivery. EmpowerTours SAS de CV is incorporated in Mexico.

What we do not have, plainly:

- no cryptographer on staff — which is why a third-party audit is item **1** of the funding plan, not item 6
- no institutional sales experience — the design-partner phase exists to buy that knowledge, not to fake it
- no dedicated security engineer for HSM key custody

> *Two people shipped four verified mainnet contracts, a ZK circuit, and 448 passing tests. That is the argument for a pilot — and the reason the ask is a counterparty, not headcount.*

WHAT'S NEXT

From live mainnet proof to institutional pilot.

Now → Q3 2026

- Independent third-party security audit (contracts + PQ/ZK)
- Package as a B2B SDK (wallets, custodians, bridges)
- HSM/KMS-backed key custody

Q4 2026

- First institutional design-partner pilot
- Hot-path PQ cosigner service
- Paid bug bounty (Immunefi / Code4rena)

THE ASK

A 12-week paid pilot.
Not a cheque.

One asset flow. Agreed pass/fail criteria. $75-120k — sized to fund a scoped audit of the settlement path, not the full-stack engagement.

We will have failed if any settlement instruction cannot be tied on-chain to the exact PQ-signed order that authorised it, within the agreed gas and latency budget — and we will report it that way.

Our binding constraint is not capital. The mainnet deploy cost trivial gas and is already done.

It is the absence of an institutional counterparty willing to define what good looks like.

`github.com/EmpowerTours/quantum-portfolio` · commit `9727322`